

Lab 1. Environment Setup & Java Basics

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2.2.1 Write, compile the first Java application:

Code:

```
1 public class Ex_2_2_1 {  
2     Run | Debug | Run main | Debug main  
3     public static void main(String args[]) {  
4         System.out.println(x: "Xin chao \n cac ban!");  
5         System.out.println(x: "Hello \t world!");  
6     }  
7  
8 }
```

Result:

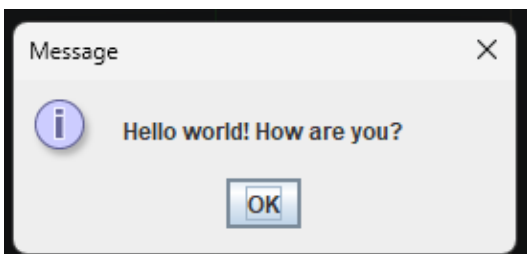
```
PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 1  
Listening on 63198  
User program running  
Xin chao  
cac ban!  
Hello world!  
User program finished
```

2.2.2 Write, compile the first dialog Java program

Code:

```
1 import javax.swing.JOptionPane;  
2  
3 public class Ex_2_2_2 {  
4     Run | Debug | Run main | Debug main  
5     public static void main(String[] args) {  
6         JOptionPane.showMessageDialog(parentComponent: null, message: "Hello world! How are you?");  
7         System.exit(status: 0);  
8     }  
9 }
```

Result:

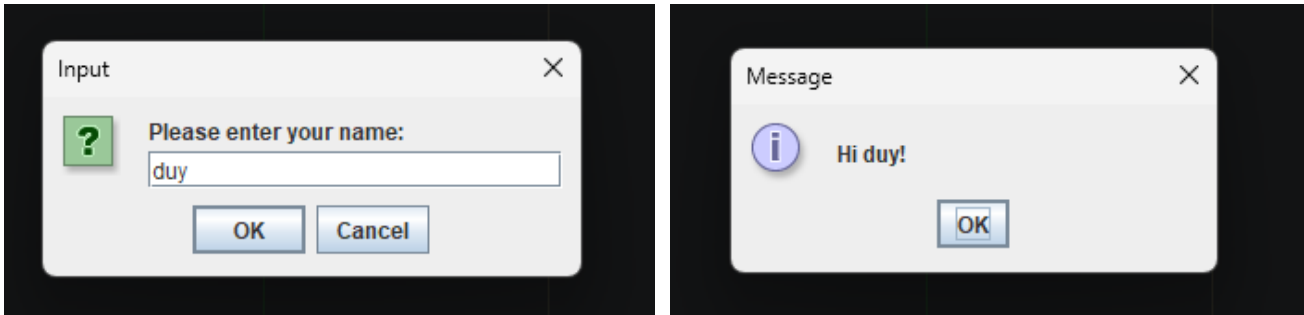


2.2.3 Write, compile the first input dialog Java application

Code:

```
1 import javax.swing.JOptionPane;
2
3 public class Ex_2_2_3 {
4     Run | Debug | Run main | Debug main
5     public static void main(String[] args) {
6         String result;
7         result = JOptionPane.showInputDialog(message: "Please enter your name:");
8         JOptionPane.showMessageDialog(parentComponent: null, "Hi " + result + "!");
9         System.exit(status: 0);
10    }
11 }
```

Result:

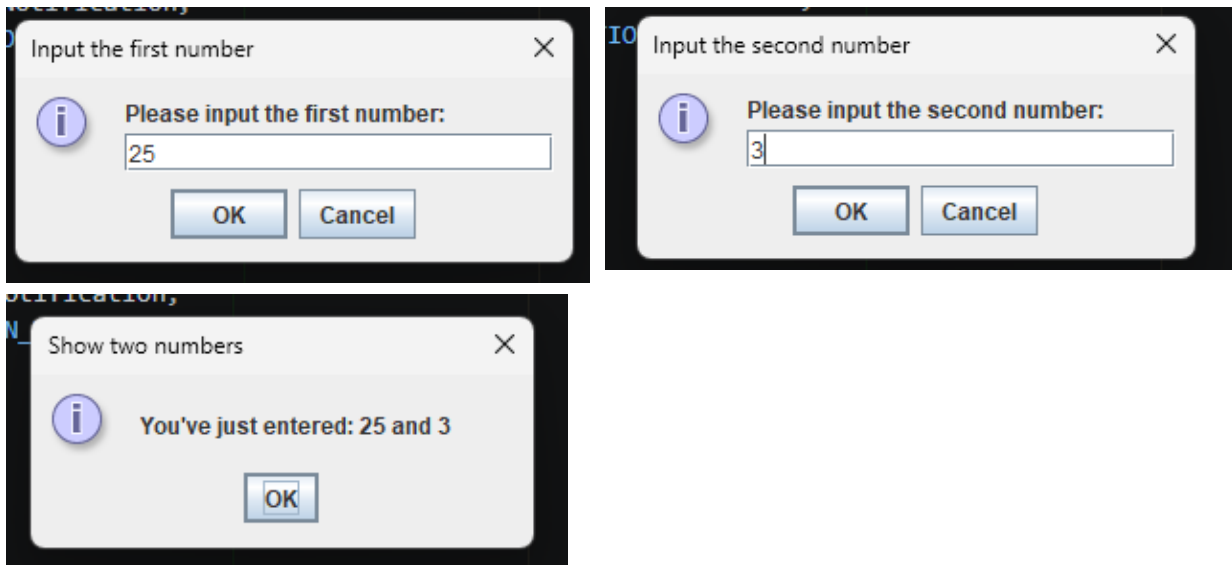


2.2.4 Write, compile, and run the following example:

Code:

```
1 import javax.swing.JOptionPane;
2
3 public class Ex_2_2_4 {
4     Run | Debug | Run main | Debug main
5     public static void main(String[] args) {
6         String strNum1, strNum2;
7         String strNotification = "You've just entered: ";
8
9         strNum1 = JOptionPane.showInputDialog(parentComponent: null,
10         message: "Please input the first number: ", title: "Input the first number",
11         JOptionPane.INFORMATION_MESSAGE);
12         strNotification += strNum1 + " and ";
13
14         strNum2 = JOptionPane.showInputDialog(parentComponent: null,
15         message: "Please input the second number: ", title: "Input the second number",
16         JOptionPane.INFORMATION_MESSAGE);
17         strNotification += strNum2;
18
19         JOptionPane.showMessageDialog(parentComponent: null, strNotification,
20         title: "Show two numbers", JOptionPane.INFORMATION_MESSAGE);
21         System.exit(status: 0);
22    }
23 }
```

Result:



2.2.5 Write a program to calculate sum, difference, product, and quotient of 2 double numbers which are entered by users.

Code:

```
1  import java.util.*;
2
3  public class Ex_2_2_5 {
4      Run | Debug | Run main | Debug main
5      public static void main(String[] args) {
6          Scanner scanner = new Scanner(System.in);
7
8          System.out.print(s: "Enter first number: ");
9          double a = scanner.nextDouble();
10         System.out.print(s: "Enter second number: ");
11         double b = scanner.nextDouble();
12
13         double sum = a + b;
14         System.out.println("The sum is " + sum);
15
16         double diff = Math.abs(a - b);
17         System.out.println("The difference is " + diff);
18
19         double prod = a * b;
20         System.out.println("The product is " + prod);
21
22         String x = "The sum is " + sum + "\nThe difference is " + diff +
23         "\nThe product is " + prod;
24
25         if (b != 0) {
26             double quot = a / b;
27             x += "\nThe quotient is " + quot;
28         } else {
29             x += "\nCannot divide by zero!!!";
30         }
31         System.out.println(x);
32         scanner.close();
33     }
```

Result:

```
Enter first number:
→ 5

Enter second number:
→ 4

The sum is 9.0
The difference is 1.0
The product is 20.0
The sum is 9.0
The difference is 1.0
The product is 20.0
The quotient is 1.25
```

2.2.6 Write a program to solve the first-degree equation, the system of first-degree equations with two variables, and the second-degree equation with one variable.

Code:

```
1 import java.util.*;
2
3 public class Ex_2_2_6 {
4     Run | Debug | Run main | Debug main
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7
8         int deg = 0;
9
10        while (true) {
11            System.out.print(s: "Enter the degree of equation (1 or 2): ");
12            int input1 = sc.nextInt();
13            if (input1 == 1 || input1 == 2) {
14                deg = input1;
15                break;
16            } else {
17                System.out.println(x: "Invalid input, try again.");
18            }
19        }
20
21        int variable = 0;
22
23        while (true) {
24            if (deg == 1) {
25                System.out.print(s: "Enter the number of variables (1 or 2): ");
26                int input2 = sc.nextInt();
27                if (input2 == 1 || input2 == 2) {
28                    variable = input2;
29                    break;
30                } else {
31                    System.out.println(x: "Invalid input, try again.");
32                }
33            } else {
34                variable = 1;
35                break;
36            }
37        }
38
39        System.out.println("Degree selected: " + deg);
40        System.out.println("Number of variables: " + variable);
41
42        // Case 1: Linear, 1 variable
43        if (deg == 1 && variable == 1) {
44            int a;
45            while (true) {
46                System.out.print(s: "Enter a (for ax + b = 0): ");
47                int a0 = sc.nextInt();
48                if (a0 != 0) {
49                    a = a0;
50                    break;
51                } else {
52                    System.out.println(x: "Invalid input, a must differ from 0, try again.");
53                }
54            }
55        }
56    }
57 }
```

```
55     System.out.print(s: "Enter b (for ax + b = 0): ");
56     int b = sc.nextInt();
57     double result = -(double)b / a;
58     System.out.println("The solution is x = " + result);
59 }
60
61 // Case 2: Linear, 2 variables
62 if (deg == 1 && variable == 2) {
63     System.out.print(s: "Enter a11 (for a11*x1 + a12*x2 = b1): ");
64     int a11 = sc.nextInt();
65     System.out.print(s: "Enter a12 (for a11*x1 + a12*x2 = b1): ");
66     int a12 = sc.nextInt();
67     System.out.print(s: "Enter b1 (for a11*x1 + a12*x2 = b1): ");
68     int b1 = sc.nextInt();
69     System.out.print(s: "Enter a21 (for a21*x1 + a22*x2 = b2): ");
70     int a21 = sc.nextInt();
71     System.out.print(s: "Enter a22 (for a21*x1 + a22*x2 = b2): ");
72     int a22 = sc.nextInt();
73     System.out.print(s: "Enter b2 (for a21*x1 + a22*x2 = b2): ");
74     int b2 = sc.nextInt();
75
76     int D = a11 * a22 - a12 * a21;
77     int D1 = b1 * a22 - b2 * a12;
78     int D2 = a11 * b2 - a21 * b1;
79
80     if (D != 0) {
81         double x1 = (double) D1 / D;
82         double x2 = (double) D2 / D;
83         System.out.println(x: "Unique solution:");
84         System.out.println("x1 = " + x1);
85         System.out.println("x2 = " + x2);
86     } else {
87         if (D1 == 0 && D2 == 0) {
88             System.out.println(x: "The system has infinitely many solutions.");
89         } else {
90             System.out.println(x: "The system has no solution.");
91         }
92     }
93 }
94
95 // Case 3: Quadratic, 1 variable
96 if (deg == 2 && variable == 1) {
97     int a;
98     while (true) {
99         System.out.print(s: "Enter a (for a*x^2 + b*x + c = 0): ");
100         int a0 = sc.nextInt();
101         if (a0 != 0) {
102             a = a0;
103             break;
104         } else {
105             System.out.println(x: "Invalid input, a must differ from 0, try again.");
106         }
107     }
108 }
```

```
108
109     System.out.print(s: "Enter b (for  $a*x^2 + b*x + c = 0$ ): ");
110     int b = sc.nextInt();
111
112     System.out.print(s: "Enter c (for  $a*x^2 + b*x + c = 0$ ): ");
113     int c = sc.nextInt();
114
115     double delta = b * b - 4.0 * a * c;
116
117     if (delta == 0) {
118         double x = - (double)b / (2 * a);
119         System.out.println(x: "The equation has one double root:");
120         System.out.println("x = " + x);
121     } else if (delta > 0) {
122         double x1 = (-b + Math.sqrt(delta)) / (2 * a);
123         double x2 = (-b - Math.sqrt(delta)) / (2 * a);
124         System.out.println(x: "The equation has two distinct real roots:");
125         System.out.println("x1 = " + x1);
126         System.out.println("x2 = " + x2);
127     } else {
128         System.out.println(x: "The equation has no real solution.");
129     }
130 }
131
132 sc.close();
133 }
134 }
```

Result:

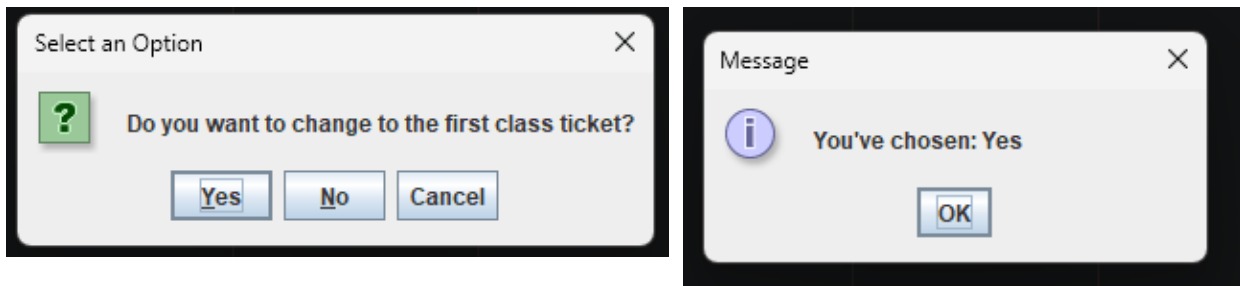
```
Enter the degree of equation (1 or 2):  
→ 1  
  
Enter the number of variables (1 or 2):  
→ 2  
  
Degree selected: 1  
Number of variables: 2  
Enter a11 (for a11*x1 + a12*x2 = b1):  
→ 3  
  
Enter a12 (for a11*x1 + a12*x2 = b1):  
→ 5  
  
Enter b1 (for a11*x1 + a12*x2 = b1):  
→ 1  
  
Enter a21 (for a21*x1 + a22*x2 = b2):  
→ 2  
  
Enter a22 (for a21*x1 + a22*x2 = b2):  
→ 3  
  
Enter b2 (for a21*x1 + a22*x2 = b2):  
→ 4  
  
Unique solution:  
x1 = 17.0  
x2 = -10.0  
User program finished
```

6.1 Write, compile and run the ChoosingOption program:

Code:

```
1 import javax.swing.JOptionPane;  
2  
3 public class Ex_6_1_ChoosingOption {  
4     Run | Debug | Run main | Debug main  
5     public static void main(String[] args) {  
6         int option = JOptionPane.showConfirmDialog(parentComponent: null,  
7             message: "Do you want to change to the first class ticket?");  
8         JOptionPane.showMessageDialog(parentComponent: null, "You've chosen: "  
9             + (option==JOptionPane.YES_OPTION?"Yes":"No"));  
10        System.exit(status: 0);  
11    }  
12 }  
13
```

Result:



6.2 Write a program for input/output from keyboard

Code:

```
1  import java.util.Scanner;
2
3  public class Ex_6_2_InputFromKeyboard {
4      public static void main(String[] args) {
5          Scanner keyboard = new Scanner(System.in);
6
7          System.out.println("What's your name?");
8          String strName = keyboard.nextLine();
9          System.out.println("How old are you?");
10         int iAge = keyboard.nextInt();
11         System.out.println("How tall are you (m)?");
12         double dHeight = keyboard.nextDouble();
13
14         System.out.println("Mrs/Ms. " + strName + ", " + iAge + " years old. "
15             + "Your height is " + dHeight + ".");
16
17         keyboard.close();
18     }
19 }
20
```

Result:

```
Listening on 57142
User program running
What's your name?
Duy

How old are you?
20

How tall are you (m)?
1.75

User program finished
Mrs/Ms. Duy, 20 years old. Your height is 1.75.
```

6.3 Write a program to display a triangle with a height of n stars (*), n is entered by users.

Code:

```
1  import java.util.*;
2
3  public class Ex_6_3 {
4      public static void main(String[] args) {
5          Scanner scanner = new Scanner(System.in);
6          System.out.print(s: "Enter the height of the triangle: ");
7          int n = scanner.nextInt();
8
9          for (int i = 1; i <= n; i++) {
10             for (int j = 1; j <= n - i; j++) {
11                 System.out.print(s: " ");
12             }
13             for (int j = 1; j <= 2 * i - 1; j++) {
14                 System.out.print(s: "*");
15             }
16             System.out.println();
17         }
18         scanner.close();
19     }
20 }
```

Result:

```
Enter the height of the triangle:
5

User program finished
*
***
*****
*****
*****
```

6.4 Write a program to display the number of days of a month, which is entered by users (both month and year). If it is an invalid month/year, ask the user to enter again.

Code:

```
1  import java.util.*;
2
3  public class Ex_6_4 {
4      Run | Debug | Run main | Debug main
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7
8          String[][] months = {
9              {"1","jan","jan.","january"},
10             {"2","feb","feb.","february"},
11             {"3","mar","mar.","march"},
12             {"4","apr","apr.","april"},
13             {"5","may","may."},
14             {"6","jun","jun.","june"},
15             {"7","jul","jul.","july"},
16             {"8","aug","aug.","august"},
17             {"9","sep","sep.","september"},
18             {"10","oct","oct.","october"},
19             {"11","nov","nov.","november"},
20             {"12","dec","dec.","december"}
21         };
22
23         int month = -1, year = -1;
24
25         while (month == -1) {
26             System.out.print(s: "Enter month: ");
27             String input = sc.nextLine().toLowerCase().trim();
28             for (String[] m : months)
29                 for (String alias : m)
30                     if (alias.equals(input))
31                         month = Integer.parseInt(m[0]);
32             if (month == -1)
33                 System.out.println(x: "Invalid month, try again.");
34         }
35
36         while (year < 0) {
37             System.out.print(s: "Enter year: ");
38             String input = sc.nextLine().trim();
39             if (input.matches(regex: "\\d+"))
40                 year = Integer.parseInt(input);
41             else
42                 System.out.println(x: "Invalid year, try again.");
43         }
44
45         int[] days = {31,28,31,30,31,30,31,31,30,31,30,31};
46         if (month == 2 && ((year%4==0 && year%100!=0) || year%400==0))
47             days[1] = 29;
48
49         System.out.println("Month " + month + " of year " + year + " has " + days[month-1] + " days.");
50         sc.close();
51     }
```

Result:

```
Enter month:
2

Enter year:
2024

User program finished
Month 2 of year 2024 has 29 days.
```

6.5 Write a Java program to sort a numeric array, and calculate the sum and average value of array elements.

Code:

```
1  import java.util.Scanner;
2
3  public class Ex_6_5 {
4      Run | Debug | Run main | Debug main
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7          try {
8              System.out.print(s: "Enter the number of elements: ");
9              int n = sc.nextInt();
10             int[] numbers = new int[n];
11             System.out.print("Enter '" + n + "' integers: ");
12             for (int i = 0; i < n; i++) {
13                 numbers[i] = sc.nextInt();
14             }
15             sortArray(numbers);
16             System.out.print(s: "Sorted array: ");
17             for (int num : numbers) {
18                 System.out.print(num + " ");
19             }
20             System.out.println();
21             System.out.println("Sum: " + sumArray(numbers));
22             System.out.println("Average: " + averArray(numbers));
23         } finally {
24             sc.close();
25         }
26     }
27
28     public static void sortArray(int[] array) {
29         for (int i = 0; i < array.length - 1; i++) {
30             for (int j = i + 1; j < array.length; j++) {
31                 if (array[i] > array[j]) {
32                     int temp = array[i];
33                     array[i] = array[j];
34                     array[j] = temp;
35                 }
36             }
37         }
38     }
39
40     public static int sumArray(int[] array) {
41         int sum = 0;
42         for (int i = 0; i < array.length; i++) {
43             sum = sum + array[i];
44         }
45         return sum;
46     }
47
48     public static int averArray(int[] array) {
49         int avg = sumArray(array) / array.length;
50         return avg;
51     }
52 }
```

Result:

```
Enter the number of elements:
3

Enter '3' integers:
2 3 5

User program finished
Sorted array: 2 3 5
Sum: 10
Average: 3
```

6.6 Write a Java program to add two matrices of the same size.

Code:

```
1  import java.util.Scanner;
2
3  public class Ex_6_6 {
4      Run | Debug | Run main | Debug main
5      public static void main(String[] args) {
6          Scanner scanner = new Scanner(System.in);
7
8          System.out.print(s: "Enter the column size of matrix: ");
9          int col = scanner.nextInt();
10         System.out.print(s: "Enter the row size of matrix: ");
11         int row = scanner.nextInt();
12
13         int[][] matrix1 = new int[row][col];
14         int[][] matrix2 = new int[row][col];
15         int[][] result = new int[row][col];
16
17         System.out.println(x: "Enter elements of first matrix:");
18         for (int i = 0; i < row; i++) {
19             for (int j = 0; j < col; j++) {
20                 matrix1[i][j] = scanner.nextInt();
21             }
22         }
23
24         System.out.println(x: "Enter elements of second matrix:");
25         for (int i = 0; i < row; i++) {
26             for (int j = 0; j < col; j++) {
27                 matrix2[i][j] = scanner.nextInt();
28             }
29         }
30
31         for (int i = 0; i < row; i++) {
32             for (int j = 0; j < col; j++) {
33                 result[i][j] = matrix1[i][j] + matrix2[i][j];
34             }
35         }
36
37         System.out.println(x: "The result is: ");
38         for (int i = 0; i < row; i++) {
39             for (int j = 0; j < col; j++) {
40                 System.out.print(result[i][j] + " ");
41             }
42             System.out.println();
43         }
44
45         scanner.close();
46     }
47 }
```

Result:

```
Enter the column size of matrix:
→ 2

Enter the row size of matrix:
→ 3

Enter elements of first matrix:
→ 1 2
→ 3 5
→
→ 2 3

Enter elements of second matrix:
→ 2 3
→ 1 4
→ 2 2

User program finished
The result is:
3 5
4 9
4 5
```